

Let know each other first.

I am Alina Calin.

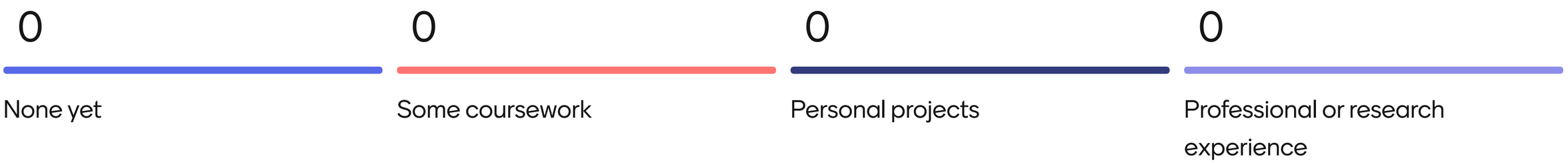


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What is one thing you enjoy building, studying, or solving?

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How much prior experience do you have with AI or machine learning?



How confident do you currently feel with these areas?

Artificial intelligence

Machine learning

Declarative programming

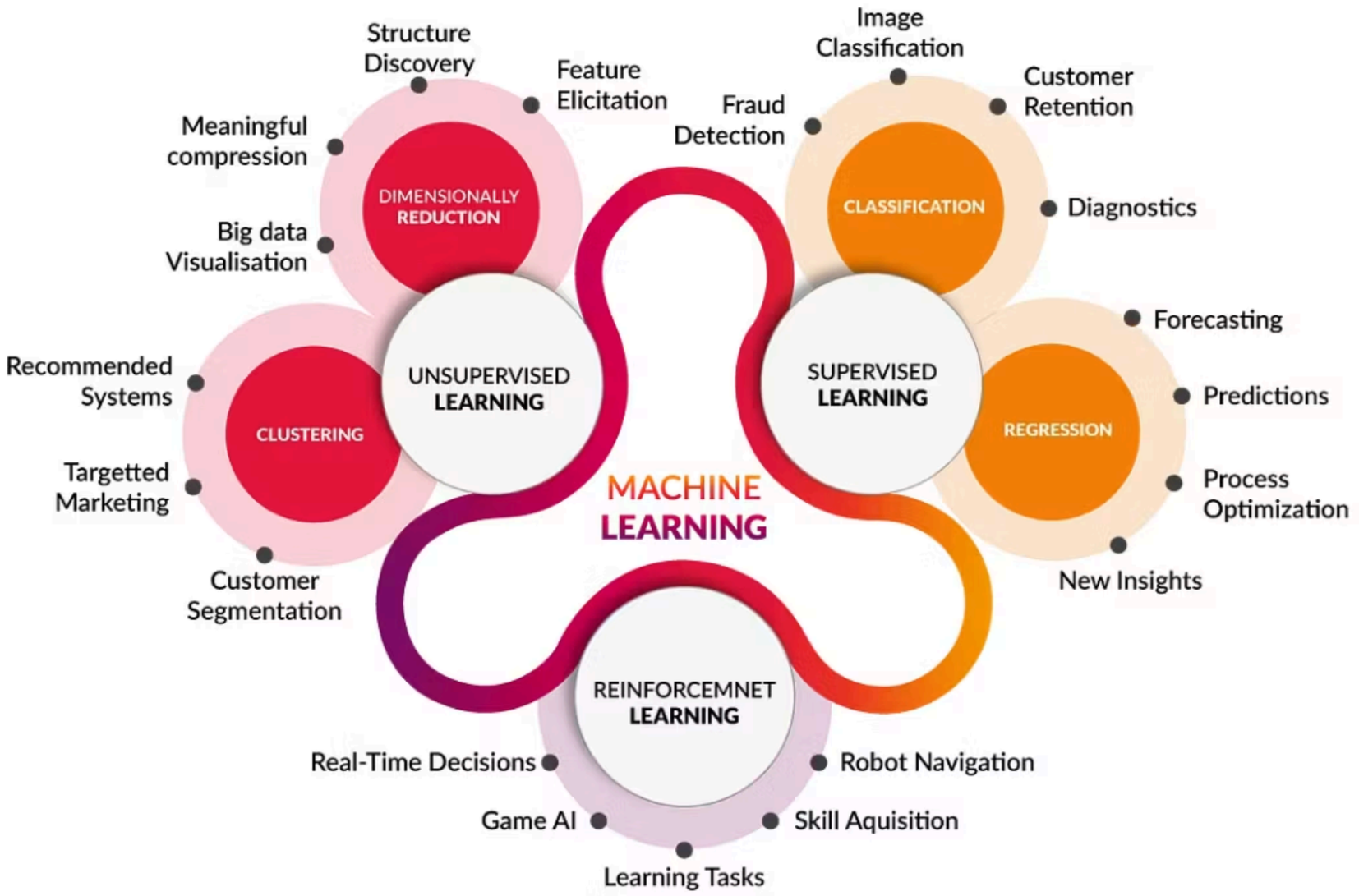
Not confident yet

Very confident

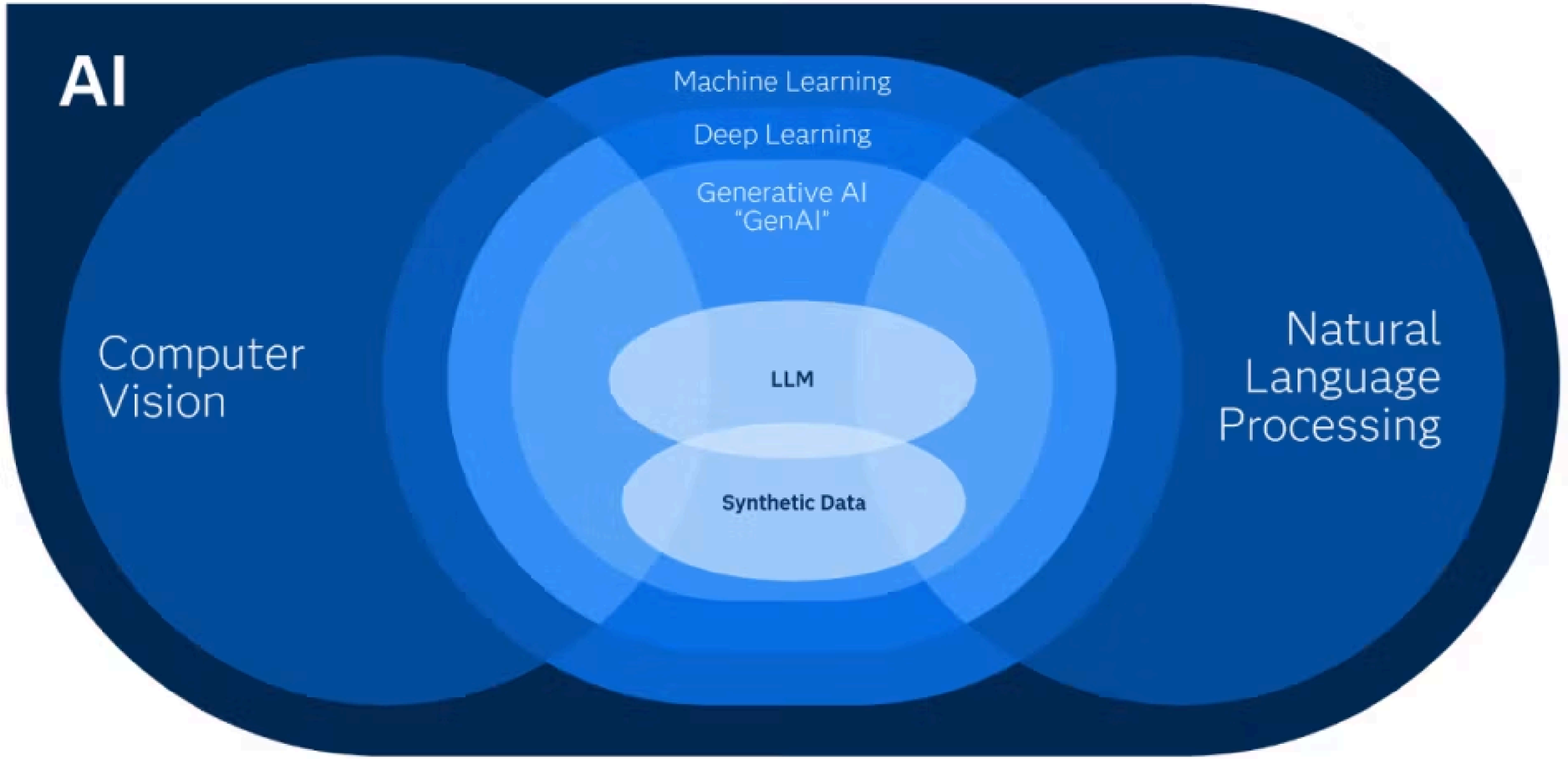
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What is your favourite AI algorithm?

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MACHINE LEARNING
UNSUPERVISED LEARNING
DIMENSIONALLY REDUCTION
Structure Discovery
Feature Elicitation
Meaningful compression
Big data Visualisation
CLUSTERING
Recommended Systems
Targetted Marketing
Customer Segmentation
SUPERVISED LEARNING
CLASSIFICATION
Image Classification
Customer Retention
Fraud Detection
Diagnostics
REGRESSION
Forecasting
Predictions
Process Optimization
New Insights
REINFORCEMNET LEARNING
Real-Time Decisions
Robot Navigation
Game AI
Skill Aquisition
Learning Tasks

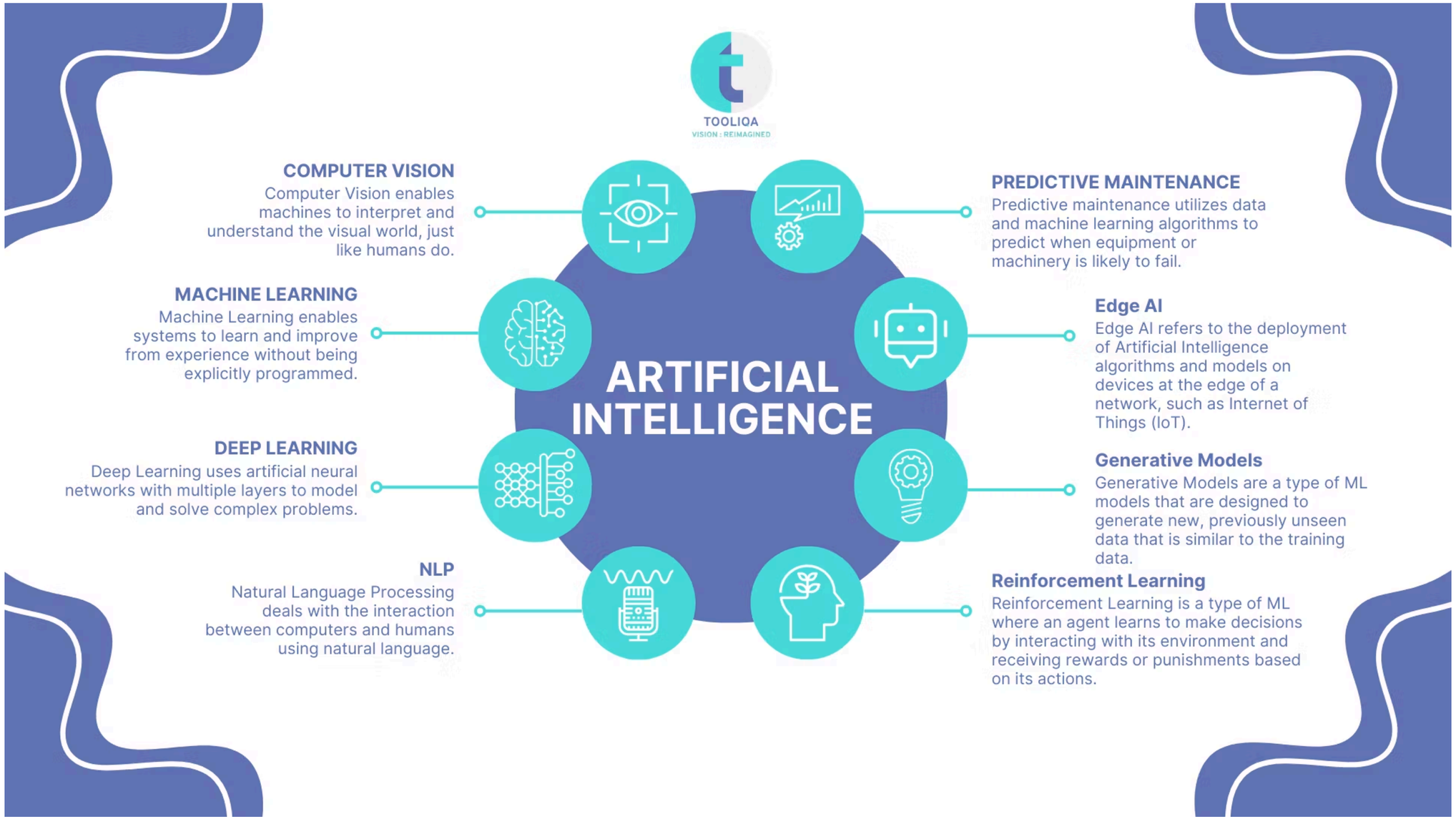
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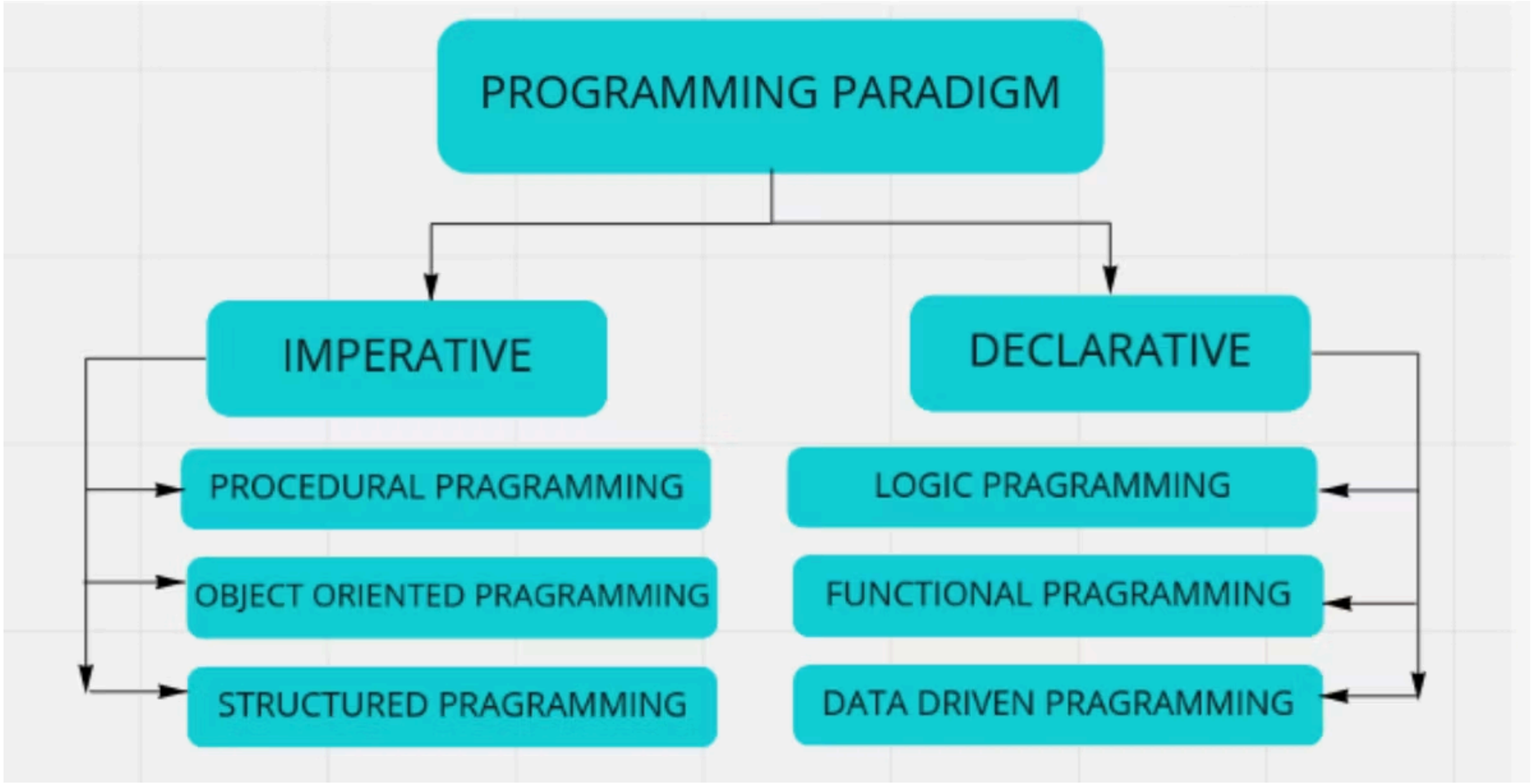
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TOOLIQA
VISION : REIMAGINED
ARTIFICIAL INTELLIGENCE
COMPUTER VISION
Computer Vision enables machines to interpret and understand the visual world, just like humans do.
MACHINE LEARNING
Machine Learning enables systems to learn and improve from experience without being explicitly programmed.
DEEP LEARNING
Deep Learning uses artificial neural networks with multiple layers to model and solve complex problems.
NLP
Natural Language Processing deals with the interaction between computers and humans using natural language.
PREDICTIVE MAINTENANCE
Predictive maintenance utilizes data and machine learning algorithms to predict when equipment or machinery is likely to fail.
Edge AI
Edge AI refers to the deployment of Artificial Intelligence algorithms and models on devices at the edge of a network, such as Internet of Things (IoT).
Generative Models
Generative Models are a type of ML models that are designed to generate new, previously unseen data that is similar to the training data.
Reinforcement Learning
Reinforcement Learning is a type of ML where an agent learns to make decisions by interacting with its environment and receiving rewards or punishments based on its actions.

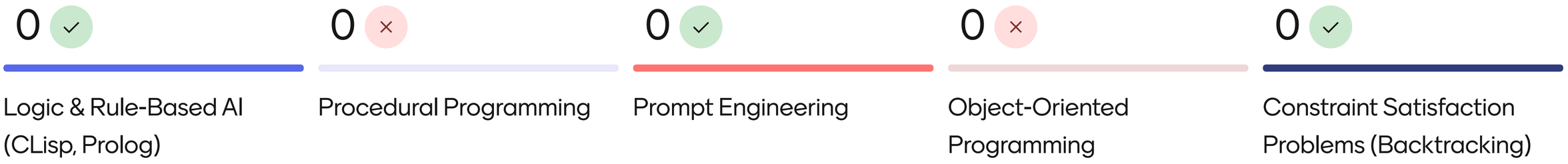
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What is one topic or challenge you would like to explore during this course?

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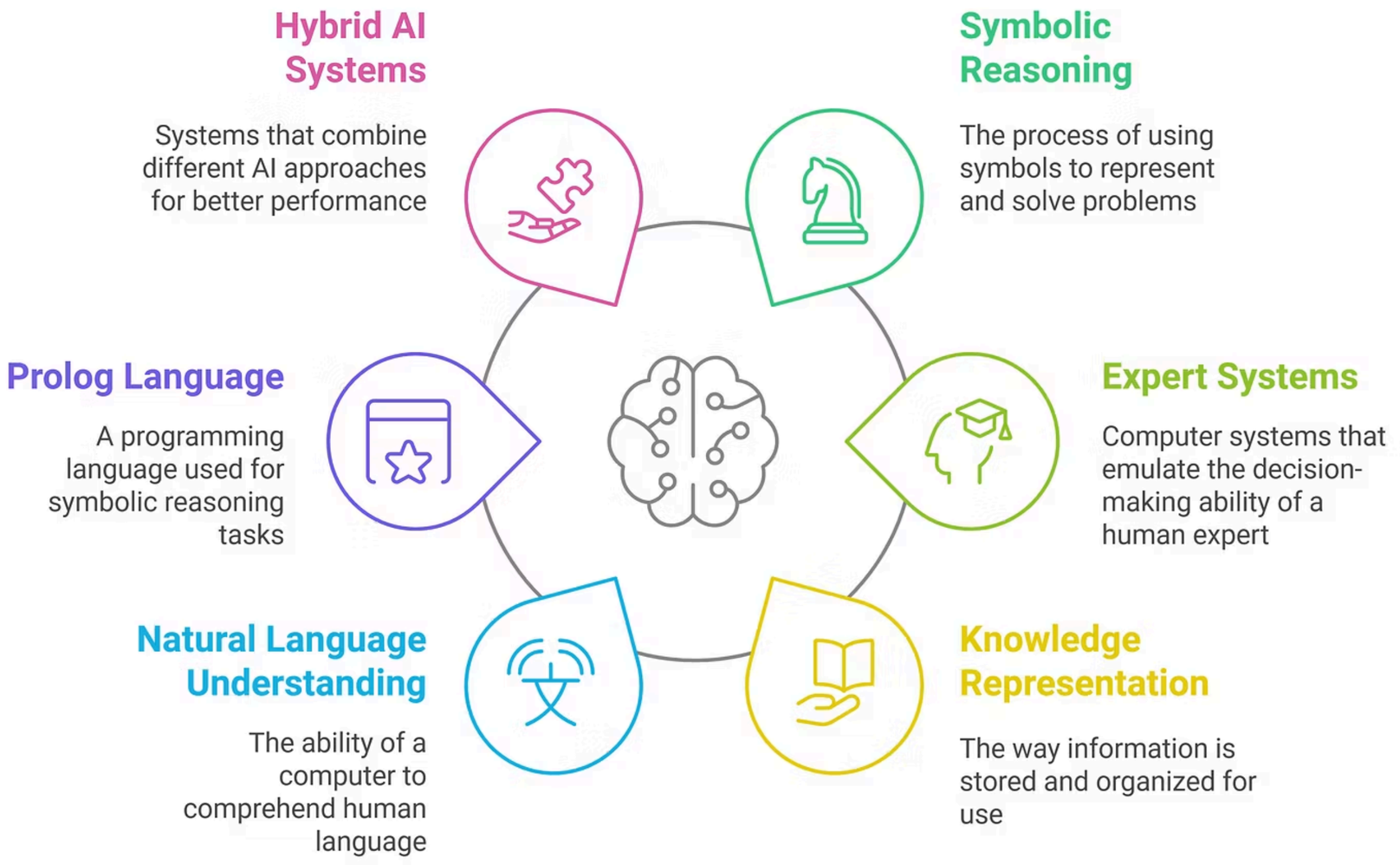

PROGRAMMING PARADIGM
IMPERATIVE
DECLARATIVE
PROCEDURAL PRAGRAMMING
OBJECT ORIENTED PRAGRAMMING
STRUCTURED PRAGRAMMING
LOGIC PRAGRAMMING
FUNCTIONAL PRAGRAMMING
DATA DRIVEN PRAGRAMMING

Which of these is declarative programming?



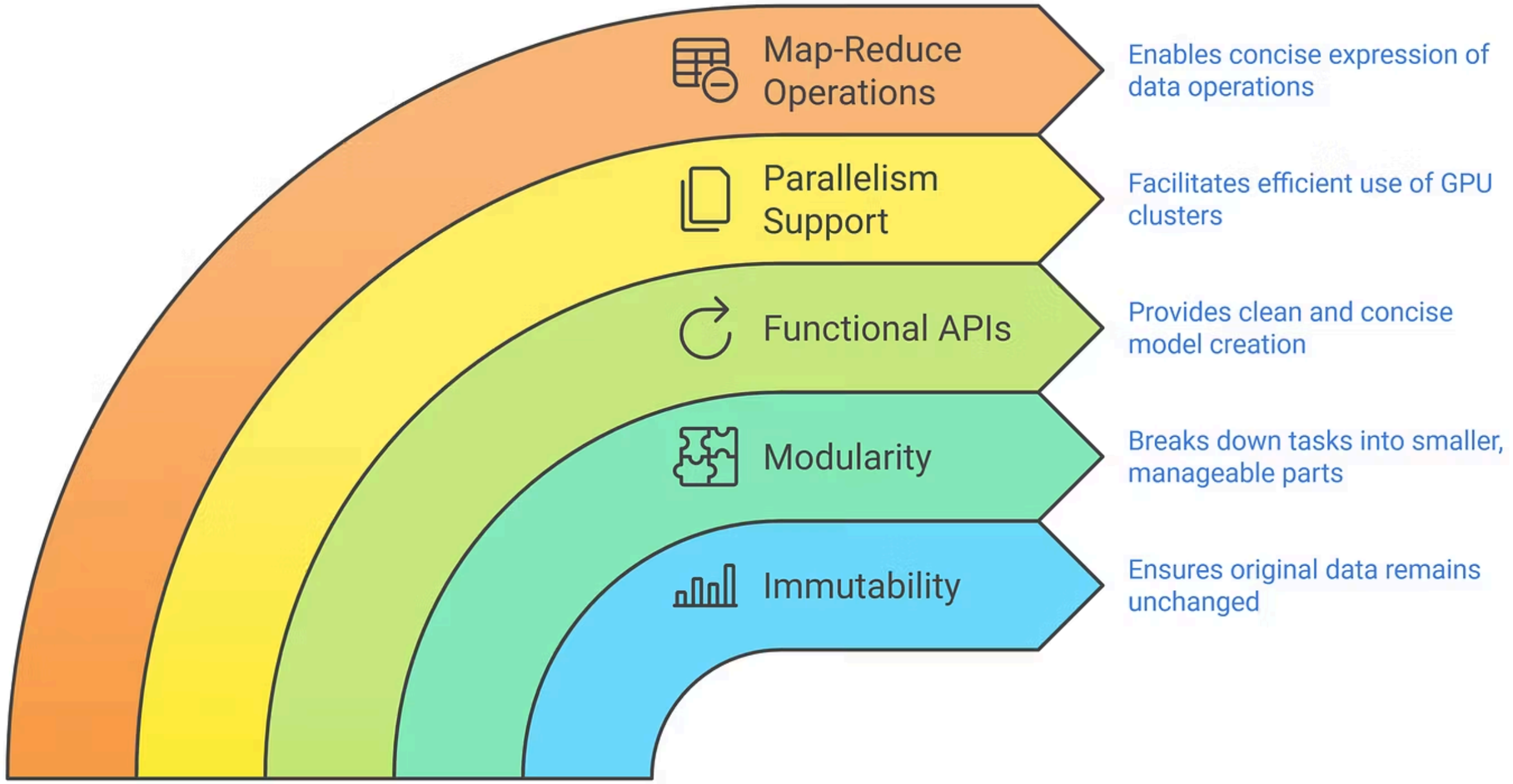
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Integration of Logic Programming in AI/ML



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Functional Programming in AI/ML



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Declarative programming

Example on CSP:

- Prolog and Constraint Satisfaction Problem (CSP) - the Zebra Puzzle (Einstain Problem)
  - **Variables (V):** The items or positions you need to determine (e.g., *House 1, House 2, House 3...*, or *Person 1, Person 2...*).
  - **Domains (D):** The possible values each variable can take (e.g., *Colors*: *Red, Green, Blue, Yellow, Ivory*; *Pets*: *Dog, Snails, Fox, Horse, Zebra*).
  - **Constraints (C):** The rules that restrict which values can be assigned to which variables simultaneously. Common Types of Constraints
    - **Unary:** Relates to a single variable (e.g., $X_1 \neq 3$).
    - **Binary:** Relates two variables (e.g., $X_1 > X_2$).
    - **Global / N-ary:** Relates multiple variables at once (e.g., $\text{alldifferent}(X_1, X_2, X_3)$).

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Real-World Examples of CSPs

Application Domain	Variables	Domain	Constraints
Sudoku	Grid cells (9×9)	Digits 1 through 9	No repeated numbers in any row, column, or 3×3 box
Scheduling	Tasks / Classes	Time slots, rooms, teachers	No double-booking of rooms or instructors; prerequisites met
Map Coloring	Map regions	Set of colors (e.g., Red, Green, Blue)	Neighboring regions cannot share the same color
N-Queens Problem	Board locations of N queens	Board coordinates	No two queens can share the same row, column, or diagonal

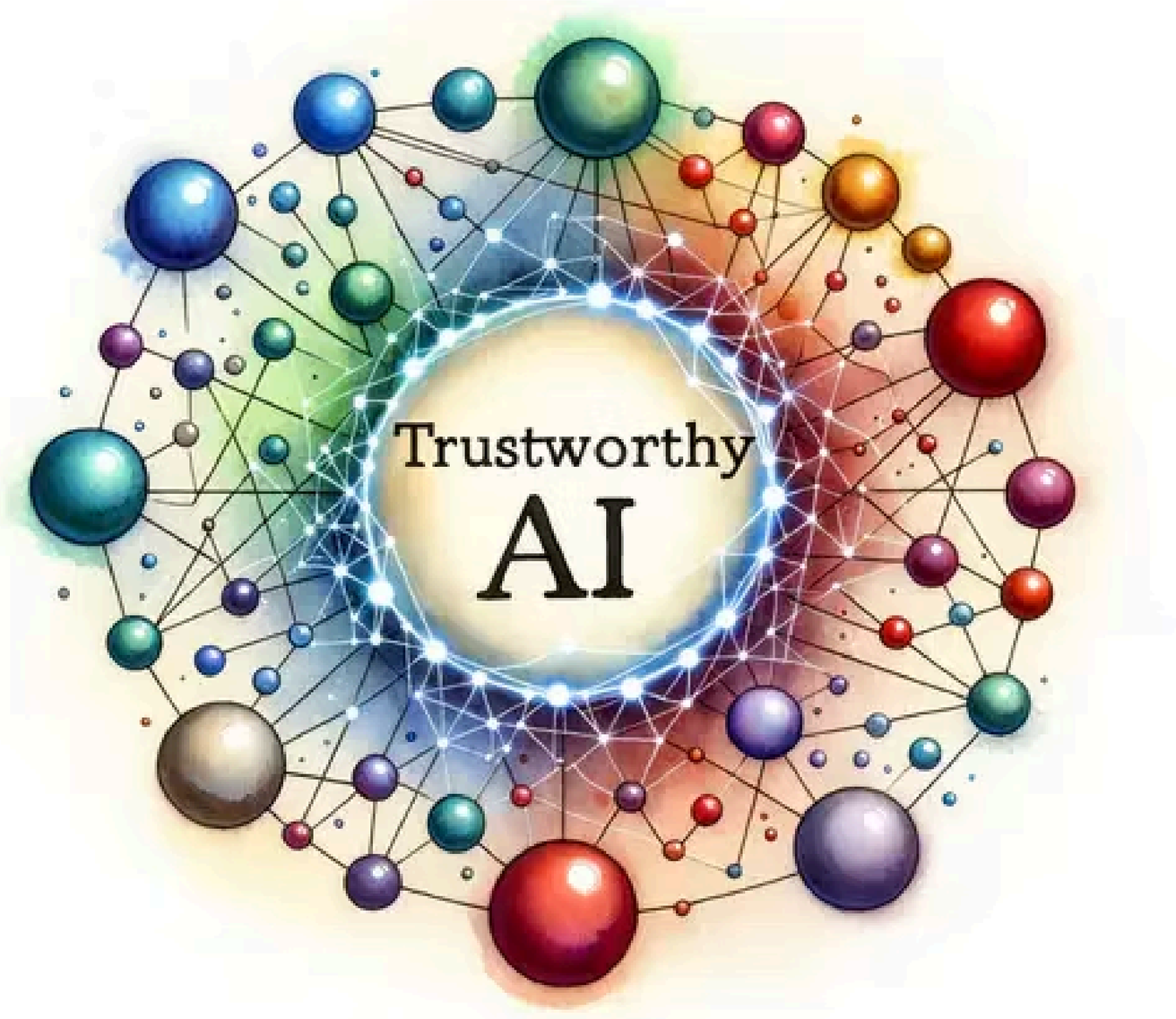
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Common Solving Techniques

Instead of using brute-force search across all potential combinations, algorithms solve CSPs by combining search with constraint propagation:

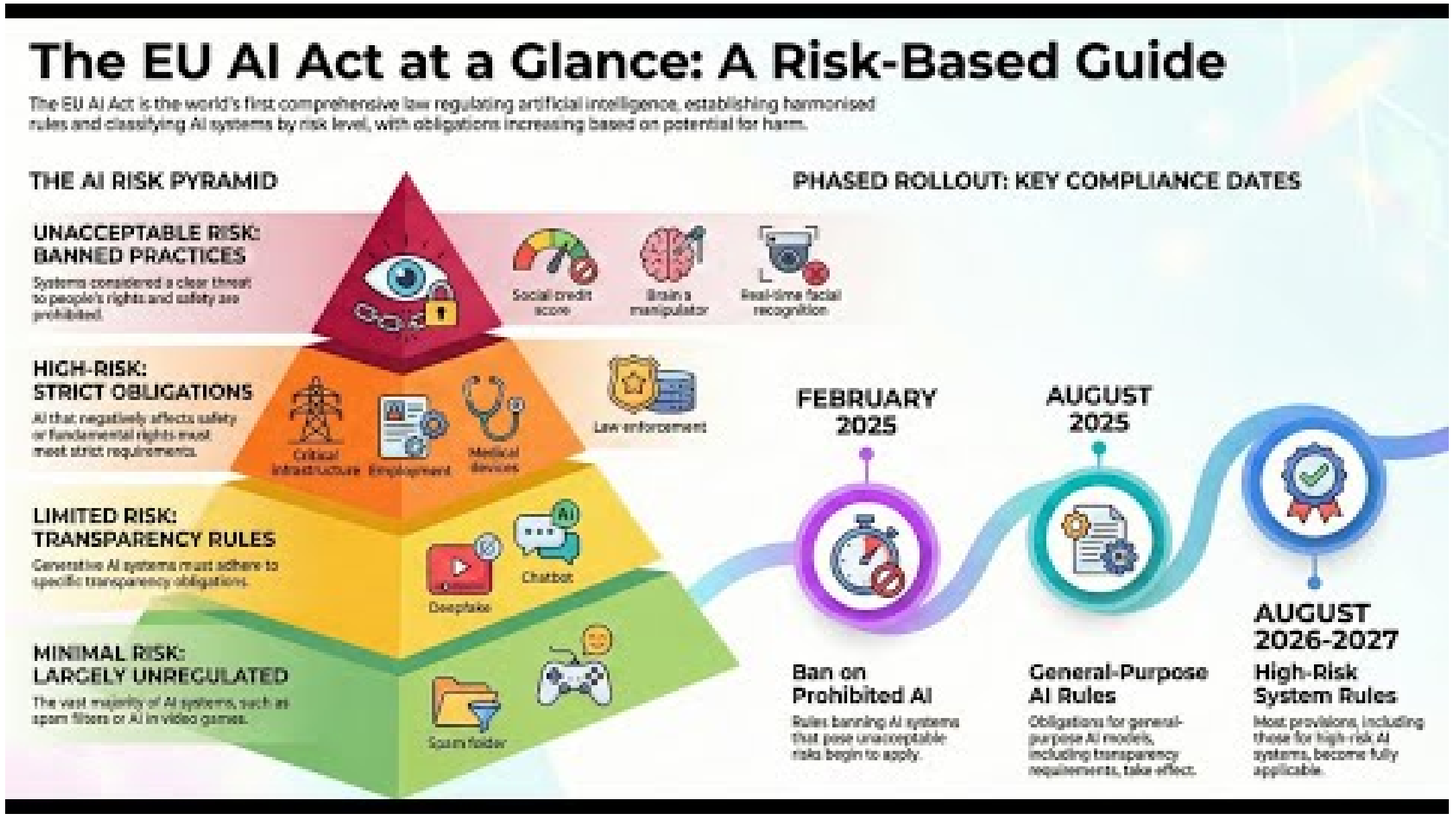
- **Backtracking Search:** A form of depth-first search that assigns values one variable at a time and backtracks as soon as a constraint is violated.
- **Arc Consistency (AC-3):** An algorithm that reduces the search space before or during search by eliminating domain values that cannot possibly satisfy binary constraints.
- **Forward Checking:** Tracks remaining valid values for unassigned variables and stops early if any variable runs out of valid domain options.

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What do you know about The EU Artificial Intelligence Act?


The EU AI Act at a Glance: A Risk-Based Guide
The EU AI Act is the world's first comprehensive law regulating artificial intelligence, establishing harmonised rules and classifying AI systems by risk level, with obligations increasing based on potential for harm.
THE AI RISK PYRAMID
UNACCEPTABLE RISK: BANNED PRACTICES
Systems considered a clear threat to people's rights and safety are prohibited.
Social credit score
Brain's manipulator
Real-time facial recognition
HIGH-RISK: STRICT OBLIGATIONS
AI that negatively affects safety or fundamental rights must meet strict requirements.
Critical infrastructure
Employment
Medical devices
Law enforcement
LIMITED RISK: TRANSPARENCY RULES
Generative AI systems must adhere to specific transparency obligations.
Deepfake
AI
Chatbot
MINIMAL RISK: LARGELY UNREGULATED
The vast majority of AI systems, such as spam filters or AI in video games.
Spam folder
PHASED ROLLOUT: KEY COMPLIANCE DATES
FEBRUARY 2025
Ban on Prohibited AI
Rules banning AI systems that pose unacceptable risks begin to apply.
AUGUST 2025
General-Purpose AI Rules
Obligations for general-purpose AI models, including transparency requirements, take effect.
AUGUST 2026-2027
High-Risk System Rules
Most provisions, including those for high-risk AI systems, become fully applicable.

Course requirements

Online